Lantern: Conflict-Aware Gradient Blending for Physics-Guided Diffusion Models in Calorimeter Simulation

Farzana Yasmin Ahmad✉
fa7sa@virginia.edu
Computer Science Department, University of Virginia
USA

Vanamala Venkataswamy
vv3xu@virginia.edu
Visiting Scholar, University of Virginia
USA

Geoffrey Fox
gcfexchange@gmail.com
Computer Science Department, Biocomplexity Institute, University of Virginia
USA

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Abstract

Monte Carlo simulation of calorimeter showers is a principal computational bottleneck for the High-Luminosity LHC, and diffusion models have emerged as fast, high-fidelity surrogates. Their denoising objective is purely statistical, however: a model can minimize it while placing the physics wrong. Existing physics-informed generative methods cannot close this gap, because they assume a closed-form law, a governing PDE residual or a hard per-sample constraint, that a shower does not supply: no per-sample PDE governs a stochastic cascade, and energy conservation fixes only one scalar per shower. Standard metrics share a blind spot of their own, comparing showers in a physics feature space while ignoring the correlation structure across calorimeter layers and voxels. We address both gaps. We introduce the Correlation Frobenius Distance (CFD), a single normalized score for correlation fidelity at layer-wise and voxel-wise scales. We then encode the soft per-sample structure available in a shower as two physics-aware auxiliary losses. The first is a variance-stabilized voxel residual loss grounded in counting statistics; the second is a graph Laplacian loss over the detector geometry. We combine both with denoising through GradBlend, which anchors the step magnitude to the denoising gradient while letting the auxiliary steer its direction, yielding Lantern, a physics-guided diffusion surrogate. On Dataset 2 of the CaloChallenge, injecting the physics losses through task-symmetric rules such as PCGrad, GradNorm, IMTL-G, and ConFIG inflates FPD by 2–100× relative to denoising alone, whereas GradBlend admits the same signal without regression and, with the Laplacian loss, Lantern improves both FPD and CFD. Our ablation study on the auxiliary loss scheduler shows that the voxel residual loss, whose gradient conflicts with denoising, requires a terminal denoising-only phase to preserve shower fidelity, whereas the non-conflicting Laplacian loss is insensitive to the schedule.

CCS Concepts

• **Computing methodologies** → **Neural networks**; *Simulation types and techniques*; • **Applied computing** → Physics.

Keywords

diffusion models, calorimeter simulation, physics-informed learning, multi-task optimization, high-energy physics

1 Introduction

Calorimeter shower simulation models the particle cascade produced when a high-energy particle strikes the detector material, and underpins detector design, calibration, and event reconstruction in high-energy physics. The standard tools are Monte Carlo simulators such as Geant4 [1], accurate but computationally expensive; the upcoming High-Luminosity upgrade of the Large Hadron Collider [4] makes Geant4 alone impractical at the required data volume. The CaloChallenge benchmark [9] spurred deep generative surrogates for full simulation, many producing high-fidelity showers at a fraction of the cost.

Evaluating these models is nontrivial: a faithful surrogate must preserve the underlying physics, not only visual plausibility. The standard distributional metrics, the Fréchet and kernel physics distances (FPD and KPD) [24], compare samples in a physics feature space and detect global mismatch, but do not test whether a model reproduces correlations between calorimeter layers and voxels; the Pearson Correlation Coefficient (PCC) [2] probes these correlations but returns a full matrix rather than a single comparable score. We introduce the Correlation Frobenius Distance (CFD), a single normalized score for layer- and voxel-wise correlation fidelity.

CFD diagnoses where a surrogate departs from Geant4 after training; a complementary question is whether physics structure can be built into training itself. The denoising objective of diffusion is purely statistical, and existing physics-informed variants assume a closed-form law that a stochastic shower does not supply (Section 2). Physics-guided training has consequently entered shower generation only through coarse aggregates: the concurrent CaloTrilogy [23] penalizes layer-wise energies with a scalar-weighted sum and offers no mechanism to keep the physics term from degrading the generative objective. We address both gaps: voxel-level losses and a combination rule that protects denoising.

Although no conservation law governs a shower voxel by voxel, each sample still carries physical structure: voxel energies whose fluctuations scale with the deposited energy, and local differences between neighboring cells set by the detector geometry. We encode these as two physics-aware auxiliary losses: a variance-stabilized voxel residual loss and a graph Laplacian loss over face-sharing voxels (Section 4). Pairing either loss with denoising turns training into a multi-task problem, and existing combination rules treat the objectives as peers, none protecting denoising when an auxiliary pulls against it. Since denoising keeps the model generative, our second contribution is a gradient blending rule that admits the physics gradient while keeping denoising primary. We study these questions on Dataset 2 of the CaloChallenge [9], a high-granularity calorimeter of 6480 voxels.

Contributions. This paper delivers a metric, two losses, and a combination rule that together make physics guidance safe for

a strong generative surrogate. (1) We introduce the *Correlation Frobenius Distance* (CFD), to our knowledge, the first single-score metric for shower correlation structure that FPD and KPD miss, at layer-wise and voxel-wise scales. (2) We design two physics-aware auxiliary losses, a variance-stabilized voxel residual loss and a graph Laplacian loss, encoding soft per-sample structure that no closed-form constraint expresses. (3) We introduce GradBlend, a denoising-anchored blending rule: the auxiliary steers only the update direction while the magnitude stays anchored to denoising, keeping the generative objective primary by construction. (4) We show this anchoring is necessary: PCGrad, GradNorm, IMTL-G, and ConFIG degrade FPD by 2–100×, while GradBlend admits the physics signal and, with the Graph Laplacian loss, improves both FPD and CFD. (5) Because the auxiliary weight cancels from GradBlend's update, the schedule reduces to a simple on/off gate; we show its terminal denoising-only phase is essential for sample coverage.

2 Related Work

Diffusion-based surrogates learn a data distribution through a single denoising objective, equivalent to learning its score [21, 44]. The generality that lets diffusion fit any distribution from samples alone leaves the physics unconstrained: a surrogate can minimize the loss while misplacing the relationships that make its outputs physically valid. This has motivated writing physics directly into generative training, through strategies that calorimeter showers fit none of. PDE-residual losses [5, 6, 52], inference-time guidance and projection [11, 19, 33, 46, 47], equivariant architectures [20, 22, 45], and energy-based steering [32, 48] each presuppose a closed-form per-sample object: a governing equation, differentiable constraint, symmetry group, or tractable energy. A calorimeter shower supplies none. It is a single stochastic Geant4 realization whose physical content lives in the ensemble, and its only hard per-sample constraints, total and per-layer energy, are already imposed by construction through conditioning and preprocessing (Section 3), leaving no per-sample target to enforce. Even where a physics loss applies, adding it to a generative objective is fragile: naive weighted sums produce pathological training dynamics when gradient scales and directions disagree [35, 49]. How the physics gradient combines with the generative gradient is therefore not an implementation detail but the central obstacle.

Expressing ensemble structure as an auxiliary loss alongside denoising turns training into a two-objective gradient problem, for which multi-task learning supplies combination rules. PCGrad [51], GradNorm [10], IMTL-G [34], and ConFIG [35] treat the objectives as peers, granting each equal influence over update direction or training rate. Peer treatment suits co-equal tasks, not a setting where one objective must remain primary: splitting the direction evenly forces denoising to cede half of each update to the auxiliary, suppressing the objective that makes the model generative. A separate line treats one task as primary: gradient cosine similarity (GCS) gates the auxiliary loss by the sign or value of its cosine with the main-task gradient, with guaranteed convergence to critical points of the main task [13], and this primary-anchored gating has since shown gains in physics-informed PDE solving [50]. GradBlend belongs to this primary-anchored family but differs in mechanism: GCS and its successors modulate the auxiliary's *scalar weight* while summing raw gradients, so a large-norm auxiliary still dominates the update direction. GradBlend instead separates direction from magnitude: it blends unit directions so neither gradient wins by norm, anchors the step size to the denoising gradient alone, and gates the auxiliary's directional share by the conflict angle.

Within calorimeter simulation, surrogates almost universally optimize a data-driven objective, encoding physics through architecture rather than the training loss [3, 7, 8, 18, 26, 30, 31, 36, 37]. The one exception, CaloTrilogy [23], adds a per-layer energy constraint to a MeanFlow generator, weighted by a fixed coefficient or an adaptive Lagrange multiplier from the modified differential method of multipliers [40]. Its constraint has a per-sample target (each layer energy has a true value), so once the representation stabilizes the physics and generative gradients agree and scalar magnitude control suffices; still, its reported sensitivity to constraint scheduling shows the combination is delicate in the weakly-conflicting regime. The auxiliaries we study have no per-sample target: they encode correlation structure that per-sample marginals cannot express, whose optimum opposes denoising and places the problem in the conflicting regime. To our knowledge, no prior calorimeter surrogate has treated physics-auxiliary training as a gradient-combination problem; characterizing that conflict and resolving it with a denoising-anchored rule is our contribution.

3 Problem Formulation

We represent a calorimeter shower in physical units (MeV) as $x \in \mathbb{R}^V_{\geq 0}$ where V is the number of voxels in a shower and its preprocessed form as $\tilde{x} = \mathcal{T}(x) \in \mathbb{R}^V$, where $\mathcal{T}$ is a composition of transformation steps. The network works in the preprocessed space $\tilde{x}$, and the inverse transform $\mathcal{T}^{-1}$ maps a sample back to physical energy x when a physical quantity is needed. Any quantity with a time index or a hat (x_t, $\hat{x}_0$) is in the preprocessed space; the bare symbol x means physical energy.

We treat simulation as conditional generation. Geant4 defines the target distribution $p(x \mid c)$, where x is the shower in physical space and c is the conditioning information. Following prior work [18, 37], we factorize the generation into two stages, each conditioned on its own subset of c, and train a separate diffusion model for each. The energy network, with parameters ϕ, models $p_\phi(u \mid E_{\text{inc}})$, where E_{inc} is the incident energy of the shower and $u \in \mathbb{R}^L$ is derived from the per-layer energies of the shower across its L longitudinal layers. The shape network, with parameters θ, models $p_\theta(x \mid E_{\text{inc}}, u)$, the voxelized shower given the incident energy and u. Marginalizing over u recovers the target:

$$p(x \mid E_{\text{inc}}) = \int p_\theta(x \mid E_{\text{inc}}, u)\, p_\phi(u \mid E_{\text{inc}})\, \mathrm{d}u. \tag{1}$$

The two networks are trained independently and composed at generation time. Our auxiliary losses and gradient method act on the shape network alone; the energy network keeps the standard denoising objective. During shape-network training, u is the ground-truth value, so the network conditions on exact layer ratios, while at generation u is drawn from the energy network.

Denoising objectives. Both networks are denoising diffusion probabilistic models (DDPM) [21]. The forward process perturbs a clean

sample z_0 under a variance schedule $\beta_1, \ldots, \beta_T$; writing $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^{t} \alpha_s$, the noised state at timestep $t \in \{1, \ldots, T\}$ is

$$z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1 - \bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I). \tag{2}$$

The energy network applies Eq. (2) to the clean ratio vector u and minimizes the standard denoising loss over $\epsilon_\phi(u_t, t, E_{\text{inc}})$; it generates the ratios autoregressively following CaloDREAM [18], carries no auxiliary loss, and is detailed in Appendix B.3. All contributions of this paper act on the shape network, which applies Eq. (2) to the preprocessed shower $\tilde{x}_0$ and minimizes, with conditioning (E_{inc}, u),

$$\mathcal{L}_{\text{denoise}}(\theta) = \mathbb{E}_{\tilde{x}_0, \epsilon, t}\left[\|\epsilon - \epsilon_\theta(x_t, t, E_{\text{inc}}, u)\|_2^2 \right]. \tag{3}$$

Physics-aware auxiliary objective. The denoising loss scores the predicted noise, not the shower. To reason about the shower itself, we use the model's single-step clean estimate

$$\hat{x}_0(x_t, t) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\, \epsilon_\theta(x_t, t, E_{\text{inc}}, u)}{\sqrt{\bar{\alpha}_t}}. \tag{4}$$

We add a physics-aware auxiliary loss on this estimate,

$$\mathcal{L}_{\text{aux}}(\theta) = \mathbb{E}_{\tilde{x}_0, \epsilon, t}\left[\ell_{\text{phys}}\big(\hat{x}_0(x_t, t),\, \tilde{x}_0\big) \right], \tag{5}$$

where ℓ_{phys} is a per-sample penalty that compares the clean estimate with the true shower. We instantiate ℓ_{phys} and describe how it is combined with the denoising objective in Section 4.

The denoising and auxiliary objectives share one update on the shape network. Writing their gradients as

$$g_d = \nabla_\theta \mathcal{L}_{\text{denoise}}, \quad g_a = \nabla_\theta\big(w_{\text{aux}}(s)\, \mathcal{L}_{\text{aux}}\big) = w_{\text{aux}}(s)\, \nabla_\theta \mathcal{L}_{\text{aux}}, \tag{6}$$

where $w_{\text{aux}}(s) \geq 0$ weights the auxiliary loss at optimization step s and the denoising loss is never reweighted, we obtain a multi-task gradient problem in which g_d and g_a need not agree. Denoising is the objective that makes the network generative, so it must stay primary; an auxiliary that overrides it degrades the samples it is meant to improve. The problem is therefore twofold. First, construct an update $g_{\text{blend}}(g_d, g_a)$ that admits the physics signal while keeping denoising primary. Second, decide when during training the auxiliary should act. We address the first with GradBlend (Section 4.2) and the second with the schedule $w_{\text{aux}}(s)$ (Section 4.3).

4 Method

Our surrogate follows the factorized design of CaloDREAM [18], with the energy and shape networks of Section 3; unlike CaloDREAM's conditional flow matching, we use DDPM [21] for both. The energy network is an autoregressive transformer over the L layers: the ratio u_i for layer i is predicted from E_{inc} and the preceding layers through causal masking, matching the forward-propagating structure of a shower. The shape network, which carries all of our contributions, is a conditional Vision Transformer [12, 39] trained to predict the injected noise $\epsilon_\theta(x_t, t, E_{\text{inc}}, u)$: it patch-embeds the calorimeter volume, processes the patches with transformer blocks conditioned on t, E_{inc}, and u through affine modulation, and projects back to voxel space. The preprocessing transform $\mathcal{T}$ (Eq. (18)) is taken unchanged from CaloDREAM; full architectural details and hyperparameters are in Appendix B.

4.1 Physics-Aware Auxiliary Losses

4.1.1 Variance-Stabilized Voxel Residual Loss. Here we take ℓ_{phys} as a per-voxel residual loss. The denoising loss Eq. (3) weights every voxel equally, but not all calorimeter voxels are equally informative. In a calorimeter shower a few bright cells carry most of the energy, while many dim cells fix the sparsity pattern and the low-energy tail. Shower signals are counting processes, so the fluctuation scale of a voxel grows as the square root of its physical energy [14]. This motivates an inverse-variance weighting that standardizes each residual by an energy-dependent scale, moving the capacity from the bright core toward the dim tail.

$$r_i = \frac{\hat{x}_{0,i} - \tilde{x}_{0,i}}{\sqrt{|\tilde{x}_{0,i}| + \epsilon_{\text{vox}}}}, \qquad \ell_{\text{vox}} = \frac{1}{V} \sum_{i=1}^{V} \text{Huber}_{\delta_{\text{vox}}}(r_i), \tag{7}$$

where $i \in \{1, \ldots, V\}$ indexes voxels and $\hat{x}_{0,i}$ and $\tilde{x}_{0,i}$ are the i-th voxel of the model's clean estimate and of the Geant4 reference, with floor $\epsilon_{\text{vox}} = 10^{-6}$ and Huber transition $\delta_{\text{vox}} = 1$. The true values $\tilde{x}_{0,i}$ are detached from the computational graph, so the denominator acts as a constant per-voxel weight rather than an optimization target. The Huber penalty

$$\text{Huber}_\delta(r) = \begin{cases} \frac{1}{2} r^2 & |r| \leq \delta, \\ \delta\left(|r| - \frac{1}{2}\delta\right) & |r| > \delta, \end{cases} \tag{8}$$

stays quadratic on the bulk of residuals and linear beyond δ_{vox}, so voxels whose residuals exceed the assumed scale bound their influence on the update rather than dominating it.

4.1.2 Graph-Laplacian Loss. The variance-stabilized voxel residual loss of Section 4.1.1 is pointwise: it ignores how each voxel relates to its neighbors, so a model can match every voxel on average yet still get the local correlations wrong. The second auxiliary loss compares the predicted and reference neighborhood structure across physically adjacent voxels.

We represent the calorimeter as a graph G with one node per voxel, connected to face-sharing neighbors along the three grid axes: periodic in azimuth φ, open radially and longitudinally. Interior voxels have six neighbors; boundary voxels have fewer, except along φ, which always supplies both. The combinatorial Laplacian $L = D - A$ (adjacency A, degree D), applied to a voxel signal y, returns at each node the summed difference to its neighbors, $(Ly)_i = \sum_{j \sim i}(y_i - y_j)$, a measure of local roughness.

In graph signal processing the Laplacian acts as a high-pass filter [43]: it responds weakly where neighboring voxels agree and strongly where they differ, picking out edges rather than flat regions. A standard regularizer minimizes the Dirichlet energy $\hat{x}_0^\top L \hat{x}_0$ to drive the signal toward smoothness, but a calorimeter shower is not smooth, so flattening it would erase the structure. We therefore penalize the gap between the predicted and reference Laplacian responses,

$$\ell_{\text{lap}} = \frac{1}{\lambda_{\max}(L)^2}\, \frac{1}{V} \left\| L\hat{x}_0 - L\tilde{x}_0 \right\|_2^2, \tag{9}$$

which is zero when the model reproduces Geant4's local differences.

The prefactor $1/\lambda_{\max}(L)^2$ keeps the auxiliary gradient on a scale comparable to the denoising gradient, as the combination scheme of Section 4.2 assumes: the squared Laplacian norm inflates the loss by up to $\lambda_{\max}(L)^2$, so dividing by $\lambda_{\max}(L)^2 \approx 95.5$ ($\lambda_{\max}(L) \approx 9.77$, computed once and cached) bounds the operator gain by one. L is

a fixed sparse operator applied twice per step, adding only $O(V)$ cost and no second forward pass.

4.2 GradBlend: A Denoising-Anchored Combination

Section 3 posed the combination problem: build a single update from the denoising gradient g_d and the auxiliary gradient g_a without letting the auxiliary displace denoising. A rule that treats the two gradients as peers fails this under conflict. GradBlend separates the update into a *direction* and a *magnitude* and assigns the two gradients different roles in each: the auxiliary may steer the direction, but the magnitude is anchored to denoising alone.

Let $\hat{g}_d = g_d/\|g_d\|$ and $\hat{g}_a = g_a/\|g_a\|$ be the unit directions of the two gradients, and let

$$\theta_{\text{conf}} = \arccos(\hat{g}_d \cdot \hat{g}_a) \tag{10}$$

be the *conflict angle* between them: below 90° the objectives reinforce, above 90° they oppose. A coefficient ρ, fixed to 1 throughout, sets the auxiliary's share of a tentative blend direction,

$$\hat{g}_{\text{tent}} = \frac{\hat{g}_d + \rho\,\hat{g}_a}{\|\hat{g}_d + \rho\,\hat{g}_a\|}, \qquad \cos\varphi_{\text{align}} = \hat{g}_d \cdot \hat{g}_{\text{tent}}, \tag{11}$$

where the *alignment angle* φ_{align} measures how far the tentative blend rotates away from the denoising direction. With $\rho = 1$ the tentative blend is the bisector of the two unit directions, and the two angles are related in closed form,

$$\cos\varphi_{\text{align}} = \sqrt{\frac{1+\cos\theta_{\text{conf}}}{2}} = \cos\left(\frac{\theta_{\text{conf}}}{2}\right), \tag{12}$$

so the entire construction is parameterized by the single conflict angle θ_{conf}.

Magnitude. The step magnitude is anchored to the denoising gradient and contracts with conflict:

$$m_{\text{ref}} = \|g_d\| \cdot \max\big(\cos\varphi_{\text{align}},\, 0.05\big) = \|g_d\| \cdot \max\Big(\cos\tfrac{\theta_{\text{conf}}}{2},\, 0.05\Big). \tag{13}$$

The auxiliary enters only through the direction, never the step size: the update equals a pure denoising step when the two gradients agree and shrinks as they conflict, so the auxiliary cannot inflate the effective learning rate, and the update stays conservative under conflict. The floor of 0.05 keeps the step from collapsing under near-antiparallel gradients.

Direction. The auxiliary contributes to the direction only when the conflict is moderate. Its effective coefficient is gated by θ_{conf}: held at full weight below 120°, decayed linearly to zero between 120° and 150°, and suppressed above. The final direction is the unit-normalized ρ_{eff}-weighted blend of $\hat{g}_d$ and $\hat{g}_a$; because both inputs are unit vectors, neither gradient can dominate the direction by raw norm.

Note that the magnitude rule (13) uses the *ungated* alignment of Eq. (11), computed at the full coefficient ρ, even in regimes where the gate reduces or removes the auxiliary from the direction. This is deliberate: a large measured conflict signals that the auxiliary disagrees strongly with denoising in the current region of parameter space, and we shrink the step as a stability precaution even when the conflicting direction is discarded. It also decouples the magnitude rule from the gate thresholds, keeping the step size a smooth function of θ_{conf} alone. Earlier variants that coupled the magnitude to the auxiliary norm were unstable in training; this rule stabilized them.

Algorithm 1 GradBlend update

Require: denoising gradient g_d, auxiliary gradient g_a
Ensure: blended update g_{blend}

$\hat{g}_d \leftarrow g_d/\|g_d\|$; $\hat{g}_a \leftarrow g_a/\|g_a\|$
$\theta_{\text{conf}} \leftarrow \arccos(\hat{g}_d \cdot \hat{g}_a)$
$m_{\text{ref}} \leftarrow \|g_d\| \cdot \max(\cos(\theta_{\text{conf}}/2),\, 0.05)$ ▷ Eqs. (12)–(13)
$\rho_{\text{eff}} \leftarrow \begin{cases} 1 & \theta_{\text{conf}} \le 120^\circ \\ 1 - \frac{\theta_{\text{conf}} - 120^\circ}{30^\circ} & 120^\circ < \theta_{\text{conf}} \le 150^\circ \\ 0 & \theta_{\text{conf}} > 150^\circ \end{cases}$
$\hat{g}_{\text{dir}} \leftarrow (\hat{g}_d + \rho_{\text{eff}}\,\hat{g}_a) \,/\, \|\hat{g}_d + \rho_{\text{eff}}\,\hat{g}_a\|$
$g_{\text{blend}} \leftarrow m_{\text{ref}} \cdot \hat{g}_{\text{dir}}$
return g_{blend}

The GradBlend update. Algorithm 1 gives the full construction.[1] Relative to a pure denoising step, the squared step-size ratio is $R^2 = \max(\cos(\theta_{\text{conf}}/2), 0.05)^2$, reaching one only at perfect alignment (Appendix D.3). The 150° fallback and the denoising anchor together keep denoising primary.

4.3 Auxiliary Weight Schedule

The schedule $w_{\text{aux}}(s)$ decides when the auxiliary loss is active. Because GradBlend normalizes each gradient to unit length and anchors the step size to the denoising-gradient norm (Section 4.2), any positive value of $w_{\text{aux}}(s)$ cancels from the update. The schedule is therefore a gate: the update is pure denoising when $w_{\text{aux}}(s) = 0$, and the auxiliary enters at full unit weight otherwise. LANTERN consequently exposes no loss-weight hyperparameter; the only design choice is when the gate opens and when it closes, which we study through three gating windows in Section 5.4. We implement the gate with cosine ramps rather than a hard switch: the ramps set the scalar auxiliary loss we report, and they are what the multi-task baselines consume, since their updates scale with $w_{\text{aux}}(s)$ directly. Every method that uses the auxiliary loss receives the same schedule, so differences between methods reflect the combination rule and not the schedule.

5 Experiments

Dataset. All experiments use Dataset 2 of the Fast Calorimeter Simulation Challenge (CaloChallenge) [15], which contains electron showers simulated with GEANT4 and incident energies drawn log-uniformly between 1 GeV and 1 TeV. Each shower has $V = 6{,}480$ voxels over 45 layers, with 16 azimuthal (φ) and 9 radial bins per layer. The first file of 100,000 showers is used for training and the second as the held-out evaluation reference. Dataset 2 is a standard benchmark whose granularity is representative of modern

[1] The implementation adds one numerical guard: it also falls back to $\hat{g}_d$ if the blended direction is nearly degenerate, $\|\hat{g}_d + \rho_{\text{eff}}\,\hat{g}_a\| < 10^{-3}$. Given the gate, this cannot occur for any θ_{conf} (the norm is bounded below by 0.5 on $\theta_{\text{conf}} \le 150^\circ$), and it never triggered in our runs.

calorimeters, and our method carries over to the finer Dataset 3 [16] from CaloChallenge 2022 geometry without modification.

Experimental Setup. The surrogate is the two-stage model of Section 4: an energy network that generates the per-layer energy ratios u and a shape network that generates the voxelized shower. The two auxiliary losses and GradBlend attach to the shape network as described there. This subsection fixes the training configuration.[2]

Both networks are denoising diffusion probabilistic models [21] with 1,000 diffusion steps. Preprocessing follows the transform defined in Appendix B.1.

Training uses a single NVIDIA V100 GPU with 32 GB of memory for both networks. The energy network is trained once for 500 epochs in approximately one hour and is fixed across all configurations. The shape network trains for 800 epochs and dominates the overall training cost, taking approximately 38 hours for most gradient-based methods (ConFIG, GradBlend, IMTL-G, and PCGrad); GradNorm is somewhat more expensive at roughly 50 hours, owing to its additional per-task gradient-norm computation. All remaining hyperparameters are listed in Table 4 of Appendix B.

Three shape networks are trained independently with different random seeds. All results are reported as mean and standard deviation over the three seeds.

5.1 Evaluation Metrics

We report CFD (Section 1) at two scales: layer-wise, over per-layer energies (Appendix A.7), and voxel-wise, over the longitudinal structure between adjacent layers, defined next.

Voxel-wise CFD. Voxel-wise CFD measures how well a model reproduces the longitudinal correlation between a voxel and the voxel at the same transverse position in the following layer. For each consecutive layer pair $(i, i{+}1)$ with $i = 1, \ldots, 44$, and each transverse position (a, r) with angular bin $a = 1, \ldots, 16$ and radial bin $r = 1, \ldots, 9$, we compute the Pearson correlation across showers between the energy at (i, a, r) and at $(i{+}1, a, r)$:

$$\rho^{(i)}_{a,r} = \frac{\sum_k \left(E_{i,a,r,k} - \bar{E}_{i,a,r}\right)\left(E_{i+1,a,r,k} - \bar{E}_{i+1,a,r}\right)}{\sqrt{\sum_k \left(E_{i,a,r,k} - \bar{E}_{i,a,r}\right)^2}\,\sqrt{\sum_k \left(E_{i+1,a,r,k} - \bar{E}_{i+1,a,r}\right)^2}}, \quad (14)$$

where $E_{i,a,r,k}$ is the energy at position (a, r) in layer i for shower k, the index k runs over the K showers in the set, and $\bar{E}_{i,a,r} = \frac{1}{K}\sum_k E_{i,a,r,k}$ is the mean energy at that position across showers. The superscript (i) identifies the layer pair $(i, i{+}1)$, and the subscript (a, r) the transverse position. Collecting $\rho^{(i)}_{a,r}$ over all 44 layer pairs and all 16×9 transverse positions gives a correlation tensor $\rho \in \mathbb{R}^{44\times16\times9}$ for Geant4 (ρ^{ref}) and for the generated samples (ρ^{gen}). We evaluate only the active positions, defined as those where both distributions carry nonzero variance, since the correlation is undefined elsewhere. Voxel-wise CFD is the normalized Frobenius distance between the two correlation tensors,

$$\text{CFD}_{\text{vox}} = \frac{\left\|\rho^{\text{gen}} - \rho^{\text{ref}}\right\|_F}{\left\|\rho^{\text{ref}}\right\|_F}, \qquad \|\rho\|_F = \sqrt{\sum_{i,a,r} \left(\rho^{(i)}_{a,r}\right)^2}, \quad (15)$$

with both sums restricted to the active positions. Lower values indicate closer agreement with Geant4's longitudinal correlation structure.

Alongside CFD and FPD/KPD, we report precision, recall, density, and coverage (PRDC) [38], which separate sample quality from diversity, and the binary classifier test [28], which trains a network to distinguish generated from Geant4 showers on low-level (voxel) and high-level (physics) features; an AUC near 0.5 indicates the two are hard to separate. Protocol details for all metrics in Appendix A.

5.2 Baselines

We consider two groups of baselines. Prior surrogates compare the absolute quality of the generated showers; combination baselines isolate the gradient combination rule as the varying component.

The prior surrogates are CaloDiffusion [3], CaloScore v2 [37], CaloDREAM [18], CaloDiT [41] and CaloDiT-2 [42]. CaloDREAM uses conditional flow matching. Standard metrics come from the CaloChallenge report [27], which evaluated all submissions with the official pipeline against the same reference file used here. We compute CFD for all models with a single procedure: from the publicly released submission samples for the prior surrogates [17], and from our own trained models for the combination baselines. The exception is CaloDiT-2, which postdates the CaloChallenge evaluation: its standard metrics come from its publication [42] and its CFD is omitted (Appendix A.1).

The combination baselines hold the architecture, auxiliary losses, weight schedule, and preprocessing fixed, and vary only the gradient-combination rule. This group contains the denoising-only model, trained without auxiliary losses, and four combination methods: PCGrad [51], GradNorm [10], IMTL-G [34], and ConFIG [35]. Beyond these gradient-level rules, we also consider two basic loss-weighting schemes, weighted sum and uncertainty weighting [25], which merge both objectives into one scalar loss and backpropagate once. Acting on the summed loss rather than individual gradients, they cannot address the gradient conflict this work targets, so we defer them to Appendix D.2: plain weighted sum and unscheduled uncertainty weighting inflate FPD by orders of magnitude, and only a scheduled uncertainty variant, which reuses our weight schedule (Section 4.3), is competitive (Tables 7 and 8).

Base denotes the denoising-only model. Models that add the auxiliary losses under a combination rule are written Base+X, for example Base+PCGrad. The proposed model, Base+GradBlend, is written Lantern. The prior surrogates keep their original names.

5.3 Results and Discussion

Table 1 reports shower fidelity on Dataset 2. Among the prior surrogates, CaloDiffusion, CaloScore v2, and the original CaloDiT trail on FPD and CFD, CaloDiT severely so. CaloDiT-2 (EDM) reports the lowest FPD in the table, its single-step CD variant trading fidelity for sampling speed, but both show weaker classifier AUCs than CaloDREAM and their CFD cannot be assessed (Section 5.2). CaloDREAM, the flow-matching model our two-stage architecture adapts, attains the best CFD and classifier AUCs and serves as our reference; Lantern reaches comparable FPD and CFD with a DDPM backbone.

[2] Code is available at https://github.com/Aaheer17/lantern-kdd27.

Table 1: Shower fidelity on CaloChallenge Dataset 2. FPD and KPD are scaled by 10^3; lower is better for FPD, KPD, and $\mathrm{CFD}_{\mathrm{vox}}$, and for AUC values closer to 0.5 are better. Classifier AUC is reported for high-level (HL) and low-level (LL) features. $\mathrm{CFD}_{\mathrm{vox}}$ is the voxel-wise Correlation Frobenius Distance of Eq. (15), computed in this work for all models including the prior surrogates; the layer-wise variant is defined in Appendix A.7. Remaining standard metrics for the prior surrogates are taken from the CaloChallenge report [27]. All auxiliary-loss models are trained under schedule S3 (Section 5.4); differences reflect the combination rule alone. Trained models report mean ± std over three seeds, and the best value among the trained models per column is in bold.

Model	Loss	FPD ↓	KPD ↓	$\mathrm{CFD}_{\mathrm{vox}}$ ↓	AUC (HL)	AUC (LL)
CaloDiffusion [3]	N/A	146.9 ± 0.9	0.174 ± 0.042	0.155	0.591 ± 0.009	0.577 ± 0.004
CaloScore v2 [37]	N/A	112.5 ± 0.9	0.149 ± 0.057	0.176	0.666 ± 0.002	0.595 ± 0.003
CaloDiT [41]	N/A	1691 ± 7	11.030 ± 0.427	0.979	0.912 ± 0.002	0.984 ± 0.001
CaloDiT-2 (EDM) [42]	N/A	20.06 ± 0.74	0.089 ± 0.090	–	0.560 ± 0.005	0.594 ± 0.002
CaloDiT-2 (CD) [42]	N/A	68.83 ± 3.37	0.390 ± 0.150	–	0.695 ± 0.004	0.598 ± 0.002
CaloDREAM [18]	N/A	24.65 ± 1.04	0.023 ± 0.036	0.113	0.521 ± 0.002	0.531 ± 0.003
Base	None	30.87 ± 1.26	0.002 ± 0.001	0.122 ± 0.004	**0.531 ± 0.009**	**0.545 ± 0.013**
Base+PCGrad	Voxel	68.50 ± 44.58	0.137 ± 0.191	0.142 ± 0.008	0.566 ± 0.027	0.560 ± 0.003
Base+GradNorm	Voxel	60.27 ± 11.15	0.102 ± 0.042	0.142 ± 0.014	0.549 ± 0.021	0.563 ± 0.001
Base+IMTL-G	Voxel	406.9 ± 80.9	2.216 ± 0.538	0.192 ± 0.012	0.807 ± 0.019	0.763 ± 0.011
Base+ConFIG	Voxel	204.7 ± 31.2	0.882 ± 0.110	0.156 ± 0.009	0.636 ± 0.003	0.592 ± 0.002
Lantern	Voxel	32.84 ± 3.27	−0.0007 ± 0.0046	0.125 ± 0.005	0.538 ± 0.002	0.558 ± 0.002
Base+PCGrad	Laplacian	169.0 ± 72.8	0.640 ± 0.392	0.155 ± 0.003	0.621 ± 0.033	0.580 ± 0.021
Base+GradNorm	Laplacian	133.5 ± 51.5	0.472 ± 0.298	0.147 ± 0.011	0.620 ± 0.036	0.582 ± 0.005
Base+IMTL-G	Laplacian	3454 ± 4820	23.520 ± 34.500	0.510 ± 0.313	0.874 ± 0.106	0.890 ± 0.088
Base+ConFIG	Laplacian	262.8 ± 73.9	1.209 ± 0.341	0.172 ± 0.007	0.656 ± 0.033	0.623 ± 0.015
Lantern	Laplacian	**25.97 ± 1.44**	−0.0048 ± 0.0009	**0.120 ± 0.005**	0.534 ± 0.003	0.554 ± 0.002

Base is a strong baseline in its own right, reaching an FPD of 30.9 and a CFD of 0.1215 without any auxiliary loss. This sets a high bar: an auxiliary physics loss is useful only if it can be added without disturbing this baseline.

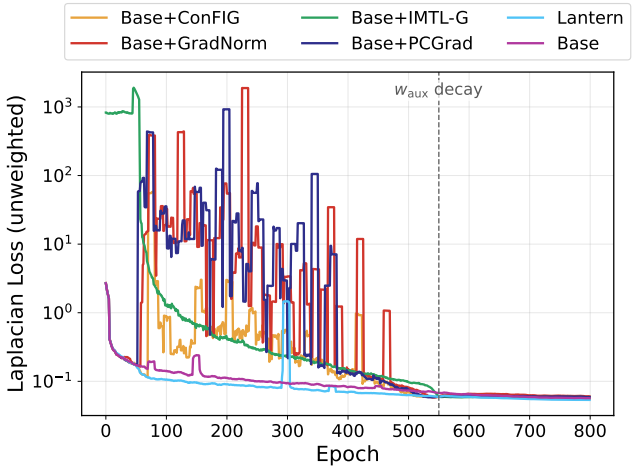
The general-purpose multi-task gradient methods do not meet that bar. Every Base+PCGrad, Base+GradNorm, Base+IMTL-G, and Base+ConFIG row has a higher FPD than Base, in several cases by more than an order of magnitude, with standard deviations large relative to the means. Base+IMTL-G is the worst, diverging on at least one seed under the Laplacian loss. These are unstable outcomes, not small regressions.

The gradient diagnostics explain the pattern. Because these methods treat the denoising and auxiliary gradients as co-equal, the combined update drifts from the denoising direction that produces valid samples, degrading fidelity; Section 5.5 reads the per-method allocations from Table 3. The training curves in Figure 1 show the resulting instability, the auxiliary loss rising by orders of magnitude and strongest under the Laplacian loss, whose high-pass gradient is noisier. Lantern is the only combination method that adds the auxiliary loss without this damage. Under the Laplacian loss it reaches the best FPD and CFD among the trained models, slightly ahead of Base; under the voxel loss it stays close to Base, with small standard deviations. Lantern integrates the auxiliary objective and preserves the fidelity of the denoising-only model. KPD does not separate the strongest models: Lantern and Base are both near zero, Lantern slightly negative. KPD is an unbiased MMD estimate that can go slightly negative when samples closely match the reference, so this is a close match, not an error, while the failing methods have large positive KPD consistent with their FPD. Classifier AUCs in the Base and Lantern rows are all near 0.5. Precision, recall, density, and coverage confirm the failure mode: the multi-task baselines gain precision and density but collapse in recall, the signature of mode collapse, whereas Base and Lantern stay balanced and close to the reference-quality prior surrogates (full breakdown in Appendix D.1, Table 5).

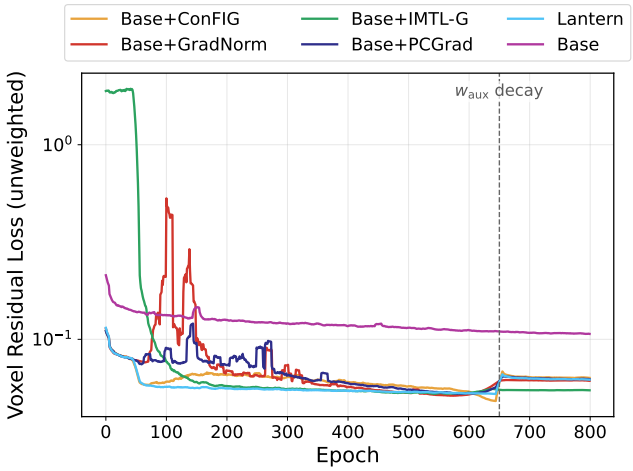
5.4 Ablation Studies

We compare three gating windows, holding every other setting fixed. Schedule S1 opens the gate at the first optimization step. S2 opens it after a 50-epoch warmup, reaching full weight by epoch 250 and holding to the end. S3 matches S2 but closes the gate before the end, decaying to zero by epoch 650 for the voxel loss and 550 for the Laplacian loss; the closing epochs were fixed a priori, not tuned. The comparison is factored: S1 versus S2 tests whether denoising must settle before the auxiliary enters, S2 versus S3 whether it must finish alone.

Timing decides voxel sample quality. Held to the last epoch (S1, S2), the voxel loss inflates FPD and density far above one while recall collapses (Table 2; full PRDC in Appendix D.1): the generator concentrates mass on part of the real distribution and fails to cover it. Coverage stays near 0.99 throughout, so this is a recall collapse, not a coverage failure. S3 reverses it, returning FPD, density, and the AUCs to near their calibrated values; precision drops correspondingly, so the terminal gate trades a little per-sample fidelity for a large gain in coverage and calibration.



(a) Laplacian loss.



(b) Voxel residual loss.

Figure 1: Unweighted auxiliary loss over training on Dataset 2. The general-purpose multi-task gradient methods produce large fluctuations, rising by several orders of magnitude at points during training. GradBlend (Lantern) keeps the auxiliary loss stable, close to the Base curve that never optimizes it. The dashed line marks the epoch at which the auxiliary weight begins to decay.

The Laplacian loss, by contrast, is insensitive to the schedule: FPD spans only 23.5 to 26.0 and CFD_{vox} 0.118 to 0.129 across S1 through S3, and its precision, density, recall, and coverage agree across schedules to within run-to-run variation (Table 6). The voxel loss competes with denoising late in training while the Laplacian loss does not, so gating it leaves the optimization trajectory and the samples unchanged. The gate is therefore necessary where the auxiliary conflicts with denoising and inert where it does not. Lantern adopts S3 for both losses, removing the voxel conflict in the final epochs at no cost to the Laplacian loss; this keeps a single schedule across auxiliaries and avoids per-loss tuning.

Table 2: Effect of the auxiliary loss schedule on Lantern.

Auxiliary	Sched.	FPD ↓	KPD ↓	CFD_{vox} ↓	AUC_{HL}	AUC_{LL}
Voxel	S1	173.4	0.997	0.123	0.777	0.783
	S2	236.3	1.528	0.138	0.833	0.838
	S3	32.84	−0.0007	0.125	0.538	0.558
Laplacian	S1	25.34	0.001	**0.118**	0.534	0.561
	S2	**23.53**	0.001	0.129	0.536	0.560
	S3	25.97	−0.0048	0.120	0.534	0.554

5.5 Training Diagnostics

Table 3 reports gradient diagnostics for the voxel energy loss, measured over the active window and averaged over three seeds. Each row summarizes how a combination method allocates its update between the denoising and auxiliary objectives. Definitions of the ratios, the active window, and the spike filter are given in Appendix D.3.

The two cosine columns separate the methods. For Base+ConFIG and Base+IMTL-G, $\cos_d$ equals $\cos_a$ in every seed. These methods weight the two gradients equally, so the resulting step makes the same angle with each. Lantern shows the widest gap between the two cosines, 0.71 against 0.21. Its step stays aligned with the denoising gradient and is held away from the auxiliary gradient. Base+PCGrad shows a partial gap, Base+GradNorm a small one.

The Balance column reads which objective each step favors. Lantern keeps the denoising objective ahead in every seed, with Balance above two. Base+GradNorm inverts this order and favors the auxiliary gradient over the denoiser, with Balance below one.

A large R_1^d is not a measure of sample quality. Base+ConFIG reaches $R_1^d = 1.42$, the highest in the table, yet has poor FPD (Table 1). The ratios describe how the step is allocated and should be read together with the evaluation tables.

The spike column reports the fraction of active epochs removed by the auxiliary-norm spike filter. Lantern has the lowest rate at eight percent, and Base+IMTL-G the highest. This ordering matches the stability seen in the training curves (Figure 1).

Figure 1 shows the unweighted auxiliary loss over training. The general-purpose multi-task methods fluctuate by orders of magnitude as the auxiliary gradient periodically overtakes the update, whereas Lantern keeps it smooth, matching the Base curve that never optimizes it while still carrying the auxiliary signal.

6 Conclusion

Physics-informed diffusion typically relies on a closed-form physical law to supervise generation, which calorimeter shower simulation lacks. We presented Lantern, which guides a diffusion surrogate with two per-sample physics losses, a variance-stabilized voxel energy residual and a graph Laplacian smoothness term, and merges their gradients with the denoising gradient through GradBlend, a combination rule that keeps denoising as the trusted reference direction while admitting each physics gradient where it agrees. These losses and GradBlend apply to the second stage, which predicts voxel energies conditioned on the first-stage incident energy and layer ratios. To measure fidelity FPD and KPD miss, we introduced CFD, the Correlation Frobenius Distance, which scores the correlation structure of calorimeter showers at two scales. On

Table 3: Gradient diagnostics over the active window, voxel energy loss on Dataset 2, reported as mean ± std over three seeds. θ_{conf} is the conflict angle between the denoising gradient g_d and the auxiliary gradient g_a; R_1^d and R_1^a are first-order progress ratios of the blended step g_{blend} on the two objectives; Balance is R_1^d/R_1^a, where a value above one favors the denoising objective; $\cos_d$ and $\cos_a$ are the cosines of g_{blend} with g_d and g_a; %spike is the fraction of active epochs removed by the auxiliary-norm spike filter. Bold marks the best value among the combination methods per column.

Method	θ_{conf}	R_1^d	R_1^a	Balance	$\cos_d$	$\cos_a$	%spike
Base+ConFIG	92.0 ± 1.2	1.42 ± 0.05	0.78 ± 0.01	1.81 ± 0.05	0.69 ± 0.01	0.69 ± 0.01	12 ± 0
Base+GradNorm	127.5 ± 3.6	0.55 ± 0.09	0.66 ± 0.07	0.83 ± 0.05	0.36 ± 0.02	0.41 ± 0.03	9 ± 0
Base+IMTL-G	116.7 ± 3.3	0.14 ± 0.01	0.10 ± 0.01	1.33 ± 0.11	0.52 ± 0.02	0.52 ± 0.02	13 ± 1
Base+PCGrad	123.9 ± 7.1	0.95 ± 0.17	0.44 ± 0.05	2.13 ± 0.20	0.55 ± 0.08	0.31 ± 0.02	11 ± 0
Lantern	119.7 ± 0.5	0.71 ± 0.02	0.24 ± 0.07	**3.26 ± 1.16**	**0.71 ± 0.02**	**0.21 ± 0.04**	**8 ± 0**

CaloChallenge Dataset 2, Lantern integrates both physics losses without the degradation symmetric multi-task rules incur: PCGrad, GradNorm, IMTL-G, and ConFIG raise FPD by factors of two to over one hundred, whereas Lantern stays close to the denoising-only baseline on FPD and KPD and remains competitive with the strongest prior surrogates on CFD and classifier AUC. We treat denoising as the primary objective; the ablations show that an auxiliary loss strengthens the model as long as its gradient stays aligned with it. The voxel gradient turns from cooperative to opposing over training, so holding it at full weight to the final epoch inflates FPD and collapses recall; a terminal phase that trains on the primary loss alone removes this and recovers both. The Laplacian gradient stays aligned throughout training and is insensitive to the schedule, so a single schedule serves both losses without per-loss tuning. We discuss limitations and next steps in Section 7.

7 Limitations and Ethical Considerations

Limitations. GradBlend and the two physics losses are demonstrated on a single controlled setting; transfer to other configurations remains future work. The geometries released with LEMURS [42], the higher-granularity Dataset 3, and hadronic showers are direct targets: adapting the method to them is an application, not a reformulation. The approach carries two sets of empirically set thresholds. The weight schedule's closing epochs, and GradBlend's angular thresholds (120° for decay and 150° for the fallback) together with the magnitude floor (0.05), were stable across the voxel and Laplacian losses here, but neither is derived, and both the schedule and the direction rule warrant further analysis, ideally tied to the measured conflict angle. We study each auxiliary loss on its own and do not characterize how the voxel-residual and Laplacian objectives interact when trained together. Results are reported over three random seeds of the shape network with the energy network fixed, so the reported variance reflects shape-network initialization alone; three seeds is common but limits power to resolve small metric differences. Physics guidance acts entirely during training and leaves the sampler unchanged, so fidelity gains come at no added generation cost. Reducing the base surrogate's sampling cost is orthogonal to our contributions and can be combined with them.

Ethical considerations. This work develops a surrogate model for calorimeter shower simulation in high-energy physics, a fundamental-physics application trained and evaluated entirely on simulated detector data from Geant4, with no human subjects, personal or sensitive data, or application to individuals. The intended benefit is reduced computational cost for physics simulation, lowering the energy and hardware demands of large-scale Monte Carlo production. Generated samples approximate rather than replace full simulation, as with any surrogate, so downstream physics analyses using them should account for the residual modeling error characterized here, particularly the correlation-structure error CFD is designed to expose. Calorimeter shower generation has no foreseeable harmful dual-use application beyond its intended scientific scope.

8 Generative AI Usage

A large language model assisted with manuscript editing and LaTeX formatting, portions of the experiment code, and cross-checking equations and related work. All contributions, including the method, losses, metric, and experimental design, are the authors' own; the authors have reviewed all AI-assisted material and take full responsibility for the paper's content, correctness, and originality.

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A Evaluation Protocol and Metric Details

A.1 Samples and Reference Data

All metrics are computed against the second file of CaloChallenge Dataset 2, which contains 100,000 Geant4 electron showers held out from training [27]. The same reference file is used for every model, including the prior surrogates we re-evaluate for CFD. For each of our trained models, we generate 100,000 showers with incident energies drawn from the reference distribution, matching the reference sample size.

For the prior surrogates, the standard metrics—FPD, KPD, PRDC, and the classifier test—are taken directly from the CaloChallenge report [27] (Tables C15, C16, and C19), which evaluated all submissions with the official pipeline against this same reference file. CFD is not reported there, so we compute it ourselves from the public submission samples released on Zenodo [17], using the identical procedure applied to our own models (Appendix A.6).

CaloDiT-2 postdates the CaloChallenge evaluation; its released artifacts include pre-trained models but no generated sample sets, so its CFD is omitted and we leave regeneration from the Geant4-distributed checkpoints to future work.

A.2 Seeds and Aggregation

The three-seed protocol and the fixed energy network are described in Section 5; the reported spread therefore reflects shape-network initialization alone. Every metric for a trained model is reported as mean ± std over the three seeds, each evaluated on its own generated sample set. Prior-surrogate rows are single values from the CaloChallenge report, carrying that report's uncertainties where available.

A.3 Standard Metric Pipeline

We compute FPD, KPD, and the classifier test with the official CaloChallenge evaluation pipeline (`evaluate.py`) [9], run unmodified, so that our trained models and the prior surrogates are scored under one protocol. Metric definitions follow the CaloChallenge report [27]. FPD and KPD are computed on the standard set of physics-motivated shower features and reported scaled by 10^3, as in the main text.

A.4 Classifier Test Protocol

The classifier test trains a binary classifier to separate generated showers (label 0) from Geant4 reference showers (label 1) and reports how well it succeeds. We evaluate two feature sets. The low-level classifier uses the voxel energies, each shower normalized by its incident energy. The high-level classifier uses the shower-shape observables of the CaloChallenge protocol [27]: total deposited energy, per-layer energies, and the energy-weighted centroids and widths in the two angular directions, together with the incident energy. Both feature sets apply the low-energy cut of 0.015 MeV used throughout our evaluation.

The classifier is the standard CaloChallenge network [27], a multilayer perceptron with two hidden layers of 512 units, trained with the pipeline's default optimizer, learning rate, and batch size. The combined samples are split into equal-sized classes and then into training, test, and validation partitions. We select the checkpoint with the best test accuracy and report the area under the ROC curve (AUC) on the held-out validation partition; an AUC near 0.5 indicates the two sample sets are indistinguishable. We report AUC averaged over three independent runs, each using samples from a separately seeded shape network, so the spread reflects variation across trained models. Prior-surrogate AUC values are taken from the CaloChallenge report, which averages ten evaluation runs [27].

A.5 PRDC Settings

Precision, recall, density, and coverage (PRDC) [38] are computed with $k = 5$ nearest neighbors, matching the CaloChallenge protocol, on the same feature space as the high-level classifier. Precision and density measure sample quality (generated samples lying on the reference manifold); recall and coverage measure diversity (reference samples reachable from the generated manifold). Prior-surrogate values are taken from Table C19 of the CaloChallenge report [27]; trained models report mean ± std over three seeds. Results are given in Table 5 in Appendix D.1.

A.6 CFD Implementation Details

CFD is computed with a single procedure for every model in Tables 1 and 5: from the public Zenodo submission samples for the prior surrogates, and from samples drawn from our own trained models otherwise. Correlations are computed on the voxel energies stored in the sample files (physical units, MeV), which already incorporate the 0.015 MeV low-energy cut; no further thresholding is applied at the correlation stage. For each model the generated and reference sets are truncated to a common size of 100,000 showers. A transverse position enters the voxel-wise computation only if it carries nonzero variance in both the reference and generated samples, since the Pearson correlation is otherwise undefined. For Dataset 2 with 100,000 showers every position satisfies this, so all $44 \times 16 \times 9 = 6{,}336$ layer-pair positions are retained and none are discarded, and both Frobenius sums in Eq. (15) run over this full set.

A.7 Layer-wise CFD

Layer-wise CFD measures the global correlation structure between per-layer energies. For each pair of layers (i, j) with $i, j = 1, \ldots, L$, we compute the Pearson correlation across showers between the

layer energies at i and j:

$$\rho_{ij} = \frac{\sum_k \left(E_{i,k} - \bar{E}_i\right)\left(E_{j,k} - \bar{E}_j\right)}{\sqrt{\sum_k \left(E_{i,k} - \bar{E}_i\right)^2}\sqrt{\sum_k \left(E_{j,k} - \bar{E}_j\right)^2}}, \tag{16}$$

where $E_{i,k}$ is the energy deposited in layer i by shower k, the index k runs over the K showers in the set, and $\bar{E}_i = \frac{1}{K}\sum_k E_{i,k}$ is the mean layer energy across showers. Collecting ρ_{ij} over all layer pairs gives an $L \times L$ correlation matrix ρ^{ref} for GEANT4 and ρ^{gen} for the generated samples. As in the voxel-wise case, we evaluate only pairs where both layers carry nonzero variance, since the correlation is undefined otherwise. Layer-wise CFD is the normalized Frobenius distance between the two matrices, restricted to the off-diagonal entries,

$$\text{CFD}_{\text{layer}} = \frac{\left\|\rho^{\text{gen}} - \rho^{\text{ref}}\right\|_{F,\,\text{off}}}{\left\|\rho^{\text{ref}}\right\|_{F,\,\text{off}}}, \qquad \left\|\rho\right\|_{F,\,\text{off}} = \sqrt{\sum_{i\neq j}\left(\rho_{ij}\right)^2}, \tag{17}$$

where both sums run over $i \neq j$. The unit diagonal is excluded because it is identical in both matrices and would only dilute the normalized distance. Lower values indicate closer agreement with GEANT4's inter-layer correlation structure. The voxel-wise form (Eq. (15), main text) applies the same construction to the per-position correlations between consecutive layers.

B Data Representation and Model Architecture

B.1 Data Representation and Preprocessing

A shower is stored as physical voxel energies $x \in \mathbb{R}^V_{\geq 0}$ in MeV, and the networks act on the preprocessed representation $\tilde{x} = \mathcal{T}(x)$ defined in Section 3. We reuse the transform of CaloDREAM [18], built from three invertible maps. Every voxel is first divided by its layer energy, so the shape network sees only the within-layer shares $s \in [0, 1]$ and the layer energies are handled separately by the ratio vector u. Each share then passes through a regularized logit,

$$s' = \log\frac{s_\alpha}{1 - s_\alpha}, \qquad s_\alpha = (1 - 2\alpha)\, s + \alpha, \qquad \alpha = 10^{-6}, \tag{18}$$

which pushes the many near-zero voxels onto a usable range while staying one-to-one, and each feature is centered and scaled using fixed training-set statistics. Generation runs $\mathcal{T}^{-1}$ and then rescales every layer's voxels to the energy set by u, fixing the total and per-layer energies of each generated shower exactly.

The ratio vector $u \in \mathbb{R}^L$ expresses the layer energies $E_0, \ldots, E_{L-1}$ through E_{inc}, following the parameterization of CaloFlow [28] adopted by CaloDREAM [18],

$$u_0 = \frac{\sum_i E_i}{f\, E_{\text{inc}}}, \qquad u_i = \frac{E_i}{\sum_{j\geq i} E_j}, \quad i = 1, \ldots, L-1, \tag{19}$$

where the constant $f = 2.85$ keeps $u_0 \in [0, 1]$. For $i > 0$, u_i is the share of the energy still undeposited at layer i that layer i receives; the map $(E_{\text{inc}}, u) \mapsto (E_0, \ldots, E_{L-1})$ therefore has a closed-form inverse and returns the layer energies without approximation.

B.2 Diffusion Background

Both networks are denoising diffusion probabilistic models [21]. The forward process noises a clean sample z_0 into z_t as in Eq. (2) of the main text, with $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s\leq t}\alpha_s$ fixed by the variance schedule $\{\beta_t\}_{t=1}^T$. Sampling inverts this process one step at a time: a noise-prediction network $\epsilon_\theta(z_t, t, c)$, conditioned on the timestep t and context c (the incident energy, and for the shape network also u), drives the ancestral update

$$z_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(z_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(z_t, t, c)\right) + \sigma_t\,\xi, \qquad \sigma_t^2 = \beta_t\,\frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}, \tag{20}$$

with $\xi \sim \mathcal{N}(0, I)$ for $t > 1$ and $\xi = 0$ at $t = 1$ for a deterministic final step. Both networks are trained with the standard noise-prediction objective

$$\mathcal{L}_{\text{denoise}} = \mathbb{E}_{z_0,\epsilon,t}\left[\,\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2\,\right]. \tag{21}$$

B.3 Energy Network: Autoregressive Transformer

The energy network predicts the layer-energy ratios $u = (u_0, \ldots, u_{L-1})$ autoregressively, reflecting that the energy reaching a layer is constrained by what has already been deposited upstream. Following the transfusion design of CaloDREAM [18], it couples a transformer encoder for the incident-energy condition with a causally masked decoder over layers and a per-layer denoising head.

Conditioning and layer inputs. The incident energy E_{inc} is embedded as a single condition token and so the encoder's self-attention runs over one element and reduces to the identity. Each layer i is marked by a positional token p_i that encodes its layer index, zero-padded to the token width; the decoder applies a causal mask, so layer i attends only to the tokens $\{p_0, \ldots, p_{i-1}\}$ that precede it. A cross-attention between the encoded condition and the decoder states produces a per-layer context c_i,

$$c_i = \begin{cases} f(E_{\text{inc}}) & i = 0, \\ f(\mathbf{p}_0, \ldots, \mathbf{p}_{i-1}, E_{\text{inc}}) & i > 0, \end{cases} \tag{22}$$

that carries what is needed to predict the ratio at layer i.

Per-layer denoising. A single denoising head ϵ_ϕ acts on the noised ratio $u_i(t)$ of each layer, conditioned on its context c_i and the timestep t, and is applied across all L layers,

$$\hat{\epsilon}\big(u(t), t, E_{\text{inc}}\big) = \big(\epsilon_\phi(u_0(t), c_0, t),\ \ldots,\ \epsilon_\phi(u_{L-1}(t), c_{L-1}, t)\big). \tag{23}$$

It is trained with the denoising objective of Eq. (21), taking $z_0 = u$. Because the clean ratios are known for every layer, training evaluates all layers in parallel; sampling instead proceeds layer by layer, each predicted ratio conditioning the next.

B.4 Shape Network: Conditional ViT

The shape network is a three-dimensional vision transformer [39], adapted to calorimeter showers as in CaloDREAM [18], that denoises the preprocessed shower conditioned on (E_{inc}, u). The calorimeter volume is split into non-overlapping voxel patches; each patch is linearly embedded, tagged with a learnable positional encoding, and processed by a stack of transformer blocks of multi-head self-attention and feed-forward layers. A final linear layer maps each patch back to voxel space as a noise estimate, and the patches are reassembled into the shower volume. It is trained with the objective of Eq. (21), taking $z_0 = \tilde{x}$; the input and patch dimensions are listed in Table 4.

Table 4: Hyperparameters of the energy and shape networks on Dataset 2.

Energy network	
Parameter	**DS2**
Epochs	500
LR sched.	one-cycle
Max LR	2×10^{-3}
Batch size	4000
Noise schedule	Linear
Solver	DDPM ancestral sampler
Conditioning network (Transformer)	
Dim embedding	64
Intermediate dim	512
Num heads	4
Encoder layers	4
Decoder layers	4
Normalization	LayerNorm
Denoising network (dense feed-forward)	
Intermediate dim	256
Num layers	8
Activation	SiLU
Training	
Optimizer	AdamW
Betas	(0.9, 0.999)
Timesteps	1000

Shape network	
Parameter	**DS2**
Epochs	800
LR sched.	CosineAnnealing
LR / Max LR	5×10^{-4} / 1×10^{-3}
Batch size	64
Noise schedule	sigmoid
Solver	DDPM ancestral sampler
Backbone network (ViT)	
Input shape	[1, 45, 16, 9]
Patch shape	[3, 16, 1]
Hidden dim	240
Depth (layers)	6
Num heads	6
MLP ratio	4.0
Condition dim	46
Attention	Standard (non-causal)
Prediction type	noise
Training	
Optimizer	AdamW
Betas	(0.9, 0.999)
Epsilon	1×10^{-6}
Weight decay	1×10^{-5}
Timesteps	1000

The conditioning inputs, the timestep t, the incident energy E_{inc}, and the ratio vector u, are embedded by small dense networks and summed into a single vector that modulates every block. The modulation is affine: the condition sets scale and shift parameters (a, b) and a residual gate γ applied to both sublayers,

$$x_h = x + \gamma_h\, g_h(a_h x + b_h), \qquad x_l = x_h + \gamma_l\, g_l(a_l x_h + b_l), \tag{24}$$

where g_h and g_l are the self-attention and feed-forward operations and (a, b, γ) depend on the conditioning vector [39]. Patch size trades spatial detail against attention cost; the value used for Dataset 2 is given in Table 4.

B.5 Training and Hyperparameters

As noted earlier, the Autoregressive Layer Network and the Shape Network are trained independently but coupled during generation. All experiments were conducted on a single NVIDIA Tesla V100-SXM2 GPU with 32 GB of memory, using CUDA 12.8 for both training and inference, with all models implemented in PyTorch. Table 4 reports the hyperparameters for both the energy and shape networks.

C Auxiliary-Loss Combination Baselines

C.1 Gradient-Combination Baselines

The combination baselines isolate the rule that merges the denoising and auxiliary gradients. Architecture, auxiliary losses, warm-up/decay schedule, preprocessing, and the AdamW optimizer are all held fixed at the Base configuration; only the combination rule changes, so any difference between these rows reflects the rule alone. At every optimizer step we backpropagate the denoising loss and each active physics loss *separately* to full-parameter gradient vectors. We index the objectives by $i = 1, \ldots, m$, writing $\mathcal{L}_i$ for the i-th loss and $g_i = \nabla_\theta \mathcal{L}_i$ for its gradient over the shape-network parameters θ; the denoising objective is fixed at $i = 1$, so $g_1 = g_d$ and the remaining g_i are the physics-loss gradients (g_a in the two-objective case). Each rule below maps the set $\{g_i\}$ to a single update that is written into the parameters before the AdamW step. Unlike the weighted-sum baseline, these methods combine the per-objective gradients directly, so the fixed per-loss weights λ_i play no role (GradNorm instead learns its own weights, described below). If any per-objective gradient is non-finite, we fall back to the pure denoising gradient g_d for that step. In our experiments two objectives participate: the denoising loss and one physics loss (voxel or Laplacian).

ConFIG.. We apply the conflict-free update of [35] through the authors' `conflictfree` library. Each step backpropagates the denoising loss and the active physics loss separately to full-parameter gradients g_d and g_a, which are passed to `ConFIG_update` (pseudo-inverse solver, `rtol` $= 10^{-4}$, default equal-weight and projection-length models). The returned direction has equal, positive projection onto both gradients—the unit bisector of g_d and g_a in our two-objective case—and is written into the parameters before the usual AdamW step. If either gradient is non-finite we fall back to the pure denoising gradient. Architecture, auxiliary loss, schedule, preprocessing, and optimizer are identical to Base, so this configuration isolates the combination rule.

PCGrad. We apply gradient surgery [51] through the conflictfree library. Given the per-objective gradients, any pair with a negative inner product is deconflicted by projecting one gradient onto the normal plane of the other; the deconflicted gradients are then

summed and written into the parameters. We log the number of conflicting pairs resolved per step. As for the other rules, a non-finite gradient triggers the pure-denoising fallback.

IMTL-G.. We use the impartial multi-task gradient rule [34] through the `IMTLGOperator` of the `conflictfree` library. It selects the combined direction whose projection onto every objective's unit gradient $\hat{u}_i = g_i/\|g_i\|$ is equal, so no single objective dominates the shared update. We verify this impartiality online by logging the projection $\hat{u}_i \cdot g_c$ of the combined update g_c onto each objective, which the rule drives to a common value. Non-finite gradients trigger the pure-denoising fallback.

GradNorm. Unlike the three conflict-based rules, GradNorm [10] rebalances gradient *magnitudes* rather than resolving directional conflict. Each objective receives a weight w_i formed as a softmax over learned logits, normalized to sum to the number of objectives and floored at 0.5, and the applied update is the weighted sum $\sum_i w_i\, g_i$ with the weights detached—equivalent to backpropagating $\sum_i w_i\, \mathcal{L}_i$. The logits are trained (Adam, learning rate 10^{-3}) so that each objective's gradient norm is pulled toward the batch-mean norm rescaled by its relative inverse training rate raised to the power $\alpha = 1.5$; the training rate is measured against a per-objective baseline $L_{0,i}$ (the mean of $\mathcal{L}_i$ over the first 100 batches), during which an equal-weight sum is used. The gradient norms driving this weight update are measured over the full set of shape-network parameters, rather than at a single last shared layer as in the original formulation. GradNorm therefore behaves like the weighted-sum baseline but with weights adapted online in place of the fixed λ_i. A non-finite gradient triggers the pure-denoising fallback.

C.2 Loss-weighting baselines

These methods reduce the objectives to one scalar loss and backpropagate it once. They were our primary investigation before we turned to gradient-level combination: because a scalar weight sets only the *magnitude* of each loss, they cannot reorient the physics gradient away from a direction that conflicts with denoising—the limitation that motivated the gradient methods above.

Weighted sum. The simplest baseline forms a fixed linear combination $\mathcal{L} = \sum_i \lambda_i \mathcal{L}_i$ and backpropagates it once. We set $\lambda_1 = 1$ for the denoising loss and a small fixed weight for the physics loss ($\lambda_{\text{lap}} = 10^{-2}$ for the Laplacian loss and $\lambda_{\text{vox}} = 10^{-4}$ for the voxel loss). These weights are constant throughout training and, unlike the other methods, are not gated by the warm-up/decay schedule. A single scalar per loss fixes both the magnitude and the direction of the auxiliary contribution, so the physics gradient is applied in full even where it opposes denoising.

Uncertainty weighting. Following the homoscedastic formulation of Kendall et al. [25], we replace the fixed weights with learned per-loss uncertainties. With a learnable log-variance $s_i = \log \sigma_i^2$ per objective, each loss enters as $\frac{1}{2} e^{-s_i} \mathcal{L}_i + \frac{1}{2} s_i$: the precision e^{-s_i} scales the loss while the $\frac{1}{2} s_i$ term regularizes the variance. The s_i are initialized to zero and trained jointly with the network by the same AdamW optimizer. We evaluate two variants. The *pure* form applies the learned weighting throughout training, as in the original method; the *scheduled* form additionally places the auxiliary term under the same warm-up/decay envelope used by the other methods, so the schedule gates *when* the physics loss participates while its relative weight remains *learned*. In both, uncertainty weighting still reduces the two objectives to a single scalar loss, so it rescales the physics gradient but cannot reorient it.

D Additional Results

D.1 PRDC

Table 5 reports the full PRDC breakdown behind the mode-collapse finding of Section 5.3, resolving the aggregate distances FPD, KPD, and CFD into separate fidelity and diversity components. Precision and density measure fidelity; recall and coverage measure diversity [29, 38]. Density is normalized so that agreement with the reference yields values near one, and values well above one indicate over-concentration in dense regions of the reference distribution rather than higher fidelity. Base and Lantern keep these balanced, with density near 1.3 and recall near 0.21, close to CaloDREAM. The general-purpose multi-task methods instead raise density and precision while lowering recall, the signature of mode collapse: samples concentrate in high-density regions and stop covering the full reference distribution. Base+IMTL-G is the most severe. Under the Laplacian loss its density reaches 167.8 while recall falls to 0.047, so its samples pile onto a small region and miss most of the reference. Base+ConFIG and Base+GradNorm show the same trend more mildly, with density above 2 and recall below 0.18. The high precision of these methods is therefore a symptom of collapse, not a sign of quality. Lantern shows no such collapse: it keeps the highest recall among the trained models, tied with Base, and a density close to that of the reference-quality prior surrogates. Among the prior surrogates CaloDiT shows the same collapse, with density 10.2 and recall 0.060, consistent with its poor FPD.

Schedule ablation. Complementing the cross-method comparison above, Table 6 reports PRDC for the three auxiliary-loss schedules of Section 5.4. The voxel loss shows the collapse signature (high density, low recall) under S1 and S2 that the terminal gate of S3 removes, while the Laplacian loss stays near density 1.3 and recall 0.21 across all three schedules.

D.2 Results for Weighted Sum and Uncertainty Weighting

Table 7 separates the three combination rules cleanly. Scheduled uncertainty weighting attains the lowest FPD, KPD, and CFD in both representations and drives the classifier AUCs toward 0.5, the point at which generated and reference showers become hard to separate. In voxel space it reaches an FPD of 115.02 and a KPD of 0.384, more than an order of magnitude below weighted-sum (2854.24, 27.127) and close to two orders below pure uncertainty weighting (8372.79, 85.592). The classifier tests agree: the scheduled model sits at AUC 0.617 and 0.571 on the high- and low-level features, whereas the unscheduled variants stay almost perfectly separable above 0.97.

The Laplacian representation narrows this gap by improving the uncertainty-weighted models rather than weighted-sum. Pure UW falls from an FPD of 8372.79 in voxel space to 237.95, and its high-level AUC drops from 0.998 to 0.628, while weighted-sum remains poor (FPD 1518.22, AUC 0.976). Scheduled uncertainty weighting

Table 5: Precision, density, recall, and coverage (PRDC) on CaloChallenge Dataset 2, computed with $k = 5$ nearest neighbors. Prior-surrogate values are taken from the CaloChallenge report [27]. Trained models report mean ± std over three seeds.

Model	Loss	Precision ↑	Density↑	Recall ↑	Coverage ↑
CaloDiffusion [3]	N/A	0.239	1.236	0.235	0.933
CaloScore v2 [37]	N/A	0.228	1.013	0.228	0.933
CaloDiT [41]	N/A	0.500	10.23	0.060	0.924
CaloDREAM [18]	N/A	0.253	1.146	0.220	0.976
Base	None	0.264 ± 0.003	1.330 ± 0.040	0.210 ± 0.002	0.984 ± 0.002
Base+PCGrad	Voxel	0.273 ± 0.013	1.592 ± 0.339	0.199 ± 0.016	0.986 ± 0.004
Base+GradNorm	Voxel	0.280 ± 0.001	1.874 ± 0.073	0.178 ± 0.002	0.994 ± 0.001
Base+IMTL-G	Voxel	0.326 ± 0.015	5.625 ± 0.741	0.178 ± 0.021	0.891 ± 0.032
Base+ConFIG	Voxel	0.304 ± 0.001	2.674 ± 0.174	0.155 ± 0.010	0.994 ± 0.001
Lantern	Voxel	0.256 ± 0.001	1.309 ± 0.052	0.216 ± 0.004	0.981 ± 0.002
Base+PCGrad	Laplacian	0.293 ± 0.017	2.230 ± 0.548	0.177 ± 0.019	0.991 ± 0.004
Base+GradNorm	Laplacian	0.293 ± 0.009	2.326 ± 0.345	0.162 ± 0.010	0.996 ± 0.001
Base+IMTL-G	Laplacian	0.306 ± 0.177	167.8 ± 179.4	0.047 ± 0.065	0.989 ± 0.016
Base+ConFIG	Laplacian	0.302 ± 0.005	2.798 ± 0.050	0.154 ± 0.013	0.994 ± 0.003
Lantern	Laplacian	0.259 ± 0.003	1.287 ± 0.036	0.213 ± 0.004	0.983 ± 0.002

Table 6: Precision, density, recall, and coverage under the auxiliary loss schedules of Table 2.

Auxiliary	Sched.	Precision	Density	Recall	Coverage
Voxel	S1	0.520	6.471	0.081	0.992
	S2	0.586	10.090	0.060	0.987
	S3	0.256	1.309	0.216	0.981
Laplacian	S1	0.266	1.342	0.210	0.983
	S2	0.256	1.255	0.217	0.982
	S3	0.259	1.287	0.213	0.983

again gives the strongest distributional scores (FPD 148.77, KPD 0.541, CFD 0.148), and the two uncertainty variants become comparable on the classifier tests.

Two observations follow. The scheduled gate is what makes uncertainty weighting usable, since without it pure UW is the worst model in voxel space on every column. CFD follows the same ordering as FPD and KPD but on a compressed scale, and it is the only metric on which the scheduled model improves over the baselines in both representations, which supports its role as a correlation-sensitive score that FPD and KPD do not capture. The low classifier AUCs for the scheduled model should be read alongside its lower PRDC precision in Table 8: it trades per-sample sharpness for a distribution close enough to Geant4 to defeat the classifier.

PRDC analysis. The density values far above unity in Table 8, such as 317 and 437 for the voxel weighted-sum and pure uncertainty models, do not indicate stronger fidelity; they show that generated samples pile into the densest region of the real manifold, where heavily overlapping neighborhood balls inflate the per-sample count. The accompanying near-zero recall (0.011 for weighted-sum voxel) confirms mode concentration, since the samples are realistic yet enclose almost none of the real distribution. Pure uncertainty weighting raises both precision and density by driving mass into these high-density regions, but its coverage drops to 0.745 because the sparse tail modes receive few generated neighbors. The scheduled variant instead restores a density near 2 and coverage above 0.99 at the cost of lower precision, a deliberate exchange of fidelity for broader mode coverage. This behavior is sharpest under the voxel representation: the Laplacian loss keeps the uncertainty models near a density of 2, whereas weighted-sum still collapses.

D.3 Gradient Diagnostic Definitions

This section defines the quantities in Table 3. All are computed from per-epoch gradient statistics logged during training: the norms of the denoising gradient g_d, the auxiliary gradient g_a, and the applied blended step g_{blend}, together with the cosines of g_{blend} with each objective and the conflict angle between g_d and g_a.

Progress ratios. For a smooth loss L and a step $\Delta\theta$, the first-order change in L is $\langle \nabla L, \Delta\theta \rangle$. We compare the applied step $\Delta\theta_a = -\eta\, g_{\text{blend}}$ against a pure step on one objective $\Delta\theta_n = -\eta\, g_k$ by the ratio of their first-order terms and of their squared sizes,

$$R_1^k = \frac{\|g_{\text{blend}}\|}{\|g_k\|}\,\cos\big(g_{\text{blend}}, g_k\big), \qquad R_2^k = \left(\frac{\|g_{\text{blend}}\|}{\|g_k\|}\right)^2, \tag{25}$$

with $k \in \{d, a\}$ for the denoising and auxiliary objectives. The first-order ratio R_1^k is the share of that objective's progress the blended step retains. It folds in both the step size and the alignment, so it falls when the step points away from g_k. The second-order ratio R_2^k is the squared step-size ratio alone. Balance is R_1^d / R_1^a; a value above one means the step retains more of the denoising progress than of the auxiliary progress.

Table 7: Model evaluation metrics. BASE+WSUM denotes weighted-sum loss balancing, BASE+PURE UW uncertainty weighting without scheduling, and BASE+SCHED UW scheduled uncertainty weighting. CFD is computed in voxel space. AUC (HL) and AUC (LL) are the high- and low-level classifier two-sample tests, where values closer to 0.5 indicate samples less distinguishable from data.

Model	Loss	FPD ($\times 10^3$) ↓	KPD ($\times 10^3$) ↓	CFD ↓	AUC (HL)	AUC (LL)
BASE+WSUM	Voxel	2854.24 ± 14.31	27.127 ± 1.531	0.273	0.991	0.979
BASE+PURE UW	Voxel	8372.79 ± 21.14	85.592 ± 2.834	0.351	0.998	0.991
BASE+SCHED UW	Voxel	115.02 ± 2.79	0.384 ± 0.167	0.145	0.617	0.571
BASE+WSUM	Laplacian	1518.22 ± 11.29	12.571 ± 0.913	0.219	0.976	0.955
BASE+PURE UW	Laplacian	237.95 ± 4.70	1.096 ± 0.291	0.181	0.628	0.639
BASE+SCHED UW	Laplacian	148.77 ± 3.35	0.541 ± 0.178	0.148	0.645	0.583

Table 8: PRDC Evaluation Metrics

Model	Loss	Precision ↑	Recall ↑	Density ↑	Coverage ↑
BASE+WSUM	Voxel	0.933	0.011	317.0	0.930
BASE+PURE UW	Voxel	0.732	0.218	437.5	0.745
BASE+SCHED UW	Voxel	0.297	0.163	2.32	0.997
BASE+WSUM	Laplacian	0.865	0.009	123.9	0.954
BASE+PURE UW	Laplacian	0.269	0.159	3.10	0.990
BASE+SCHED UW	Laplacian	0.280	0.179	2.12	0.995

Cosines and conflict angle. $\cos_d$ and $\cos_a$ are the cosines of the blended step g_{blend} with g_d and g_a. θ_{conf} is the angle between g_d and g_a before blending, in degrees.

Active window. The auxiliary weight follows a schedule that ramps from zero, holds, and decays back toward zero. Outside the held region the auxiliary gradient is near zero by design, and the ratios in Eq. 25 diverge from the small denominator. We therefore restrict all statistics to the active window, defined per epoch as those epochs where the auxiliary weight exceeds a fixed fraction of its peak value. We use a floor fraction of 0.01, which includes the ramp.

Spike filter. Within the active window the auxiliary gradient norm can transiently collapse, which inflates R_1^a and R_2^a through the small denominator. We drop active epochs where the auxiliary gradient norm falls below one quarter of its active-window median, and report ratios on the remaining epochs. The %spike column is the fraction of active epochs removed by this filter.

Aggregation. The logged statistics are heavy-tailed, so we summarize each run by the median over its active window. We then report the mean and standard deviation of these per-run medians across three seeds. The energy network is held fixed; the three seeds vary the shape network.